Efficient Best-Of-N with Radial Consensus Score

Anonymous Author(s)
Affiliation
Address
email

Abstract

Solving complex problems with large language models (LLMs) often requires selecting the best answer from multiple candidate responses (best-of-N), yet identifying the most reliable one remains challenging, particularly when correctness does not align with majority agreement. Existing approaches based on voting or probability scoring rely on surface-level signals (e.g., frequency or token-level likelihood), often missing high-quality rare answers and ignoring the geometric structure of candidate representations, where correctness may emerge as semantic consistency rather than surface agreement. To address these limitations, we introduce **Radial Consensus Score (RCS)**, a simple, efficient, and training-free method for best-of-N selection. RCS models semantic consensus by computing a weighted Fréchet mean of answer embeddings and ranking candidates by their radial distance to this center, with the weighting scheme—uniform, frequency, probability, or cosine-similarity—controlling how the center is formed. Extensive experiments across various benchmarks covering short-form QA and long-form reasoning tasks, and four open-weight models, demonstrate that RCS variants consistently outperform strong baselines by 2–7% in selection accuracy, with advantages growing as N increases. RCS also serves as an effective drop-in replacement for majority voting in multi-agent debate and exhibits strong robustness in black-box scenarios. These results highlight geometric consensus as a scalable principle for reliable answer selection beyond majority voting.

1 Introduction

Large language models (LLMs) have achieved remarkable progress across diverse tasks, yet they remain prone to generating inconsistent or incorrect outputs, particularly in open-ended or reasoning-intensive scenarios [44, 19]. A common strategy to improve reliability is best-of-N sampling: generate multiple candidate responses from the same prompt and select the highest-quality one according to a scoring mechanism [7, 41, 21]. In this framework, performance critically depends not only on the diversity of sampled candidates but, more importantly, on the effectiveness of the selection criterion.

Existing selection methods can be broadly categorized into two groups. The first relies on *external evaluators*, such as reward models or verifiers, to score each candidate [7, 26]. While effective, these approaches introduce additional computational overhead and require extra supervision. The second group uses *intrinsic signals* derived from the sampled responses themselves. The dominant paradigm in this category is self-consistency via majority voting, which assumes that correct answers appear most frequently among samples [44]. This works well when the answer space is discrete, but it struggles in settings with semantic ambiguity or when high-quality minority answers are underrepresented despite being semantically close to the consensus [4, 21]. Recent work has also proposed ranked or reasoning-aware extensions of self-consistency, such as Borda-style voting, along with theoretical analyses connecting Best-of-N to mode estimation and KL-constrained optimization [17, 42, 40, 8]. Parallel efforts explore weighted aggregation and hybrid selection strate-

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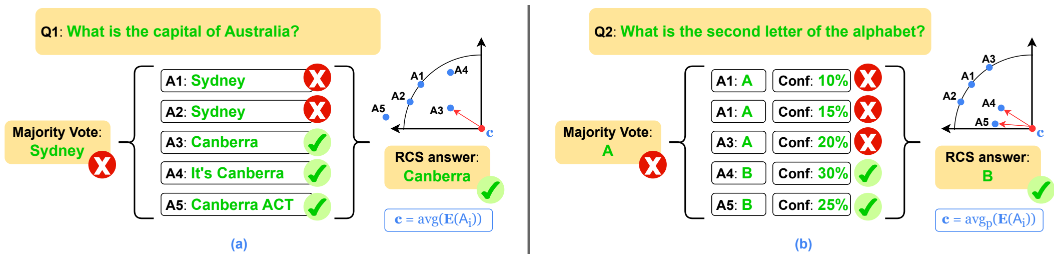

Figure 1: RCS overview and representative failure modes of majority voting. (a) Majority voting ignores semantically similar answers by treating surface forms independently. (b) It also fails to select high-confidence but minority answers. RCS addresses both issues by mapping answers into an embedding space via Encoder $\mathbf{E}$ and estimating a centroid $\mathbf{c}$, either *uniformly* or *probability-weighted* over answer embeddings. Additional qualitative examples are provided in Section 4.3.

gies, such as correlation-aware multi-agent weighting and Best-of-Majority, to improve reliability and scaling behavior [1, 10].

However, these approaches often rely on additional sampling or surface-form agreement, limiting their practicality in real-world settings, especially when efficient, black-box-compatible, and training-free selection is required. As illustrated in Figure 1, these limitations arise from a mismatch between surface-level signals and the underlying semantic structure of candidate answers. This raises a central question: *Can we develop an efficient and training-free selection framework that operates directly in answer space, enabling robust per-sample ranking without relying on external supervision or costly sampling procedures?*

To tackle this question, we introduce the **Radial Consensus Score (RCS)**, a principled geometric approach for efficient best-of-N selection. Our key insight is to model semantic consensus explicitly via a weighted Fréchet mean [13] as the semantic center of answer embeddings and then rank candidates by their radial distance to this center. This formulation naturally incorporates both semantic agreement structure and optional confidence signals (frequency, generation probability, or pairwise cosine similarity) through the choice of weighting distribution P. Different choices of P recover a spectrum of selection strategies, ranging from uniform weighting that treats all candidates equally to frequency-based, probability-based, and cosine-similarity-based schemes that emphasize consensus, model confidence, or mutual agreement. This flexibility allows the framework to adapt to different inference settings while remaining fully training-free and black-box compatible. The resulting score is computationally lightweight, model-agnostic, and directly yields a ranking of individual candidates without requiring clustering or external verifiers. We evaluate RCS on seven established benchmarks spanning short-form QA (SciQ, GPQA), mathematical reasoning (Arithmetics, GSM8K), and long-form multiple-choice tasks (MMLU Formal Logic, MMLU-Pro, BigBenchHard), using four diverse open-weight models. RCS variants consistently outperform strong baselines—including Semantic Consensus Weighting (SCW)—by 2–7% in selection accuracy, with gains becoming more pronounced as the sampling budget increases. Additional analyses confirm their efficiency and robustness in black-box settings and multi-agent debate scenarios. Our contributions are threefold: **First**, we introduce Radial Consensus Score (RCS), a geometric consensus framework for best-of-N selection that bridges probability-based and surface-form methods, and prove, under a Gaussian generative model, that it outperforms voting in both infinite- and finite-sample regimes under sufficient semantic-separation conditions. **Second**, comprehensive experiments across diverse tasks and models demonstrate that RCS consistently outperforms strong baselines, with gains increasing as the sampling budget grows, highlighting favorable scaling behavior. **Third**, ablation studies further confirm the effectiveness and applicability of RCS, showing its robustness in multi-agent debate and fully black-box settings, along with strong practical efficiency, underscoring its impact as a scalable aggregation method for LLM inference.

2 Related Work

Best-Of-N Metrics. Self-consistency, in its most common form, performs majority voting over N sampled candidates [44]. Extensions such as universal and fine-grained self-consistency improve robustness but often incur extra LLM calls for evaluation or refinement [4, 43]. Alternatively, self-certainty directly leverages the model's output probability distribution to estimate response quality without external reward models, showing strong scaling with N [21]. Beyond voting-based strategies, recent advances include ranked and reasoning-aware aggregation, e.g. Borda-style voting [42, 40]. From a theoretical perspective, Best-of-N has been analyzed under sampling and alignment objectives, linking it to mode estimation and KL-constrained optimization [17, 8]. In parallel, geometric selection methods choose candidates closest to a central representation in embedding space, with recent work leveraging embedding dispersion for uncertainty estimation [31]. Methods such as ModeX [5] instead apply spectral clustering over full reasoning traces, but are sensitive to noisy trajectories and incur higher computational cost. In contrast, our metrics operate directly at the concise answer level, yielding a lightweight and efficient framework that naturally incorporates both semantic agreement and optional confidence signals.

Several concurrent works employ embedding-space geometry for answer selection. *Semantic Consensus Weighting (SCW)* [20] embeds full reasoning trajectories, computes pairwise cosine similarities, and aggregates scores by unique reasoning path before selecting the highest-scoring group—it therefore shares the full-trajectory embedding collapse of ModeX and still relies on a discrete grouping step that inherits the minority-answer blindness of majority voting; we directly compare against SCW as a baseline (Section 4). *Centroid-based MBR* [9] selects the hypothesis nearest to the unweighted centroid to minimize expected error, but does not support confidence-weighted aggregation. *Latent Self-Consistency* [3] and ModeX [5] cluster full reasoning trajectories in latent space, incurring sensitivity to representation collapse and higher computational cost (Section 4.3). *Semantic Density* [30] measures embedding-space density as a set-level uncertainty signal rather than a per-candidate selection score. RCS unifies these directions in a single closed-form weighted Fréchet mean that operates at the final-answer level: cosine-similarity weights are used *continuously* to bias the semantic center (Table 1), bypassing any discrete grouping step and enabling recovery of minority correct answers. Formal guarantees of superiority over majority voting (Propositions 1 and 2) are absent in all cited works.

Uncertainty Estimation For LLMs. Uncertainty estimation for LLMs is typically categorized into probability-based and semantic families. (1) *Probability-based methods* rely on token-level likelihoods such as negative log-likelihood or predictive entropy [16, 28]. Recent improvements like PRO approximate entropy efficiently from the top-K probabilities with adaptive noise filtering [32]. These methods perform well on open-weight models but are inapplicable to black-box APIs and often require calibration. (2) *Semantic entropy-based methods* shift the focus to meaning by clustering responses in embedding space and computing entropy over cluster distributions or densities [22, 12, 27, 34, 35]. Extensions include evidential semantic entropy for handling uncertainty [23] and geometric dispersion measures such as semantic volume [25]. Parallel work explores token-level uncertainty through low-rank weight perturbation [48] and confidence-aware filtering of reasoning traces [14]. While these methods effectively quantify uncertainty at the set level, they largely ignore the LLM's native probability signals, are sensitive to clustering quality or computationally expensive, and require additional post-processing for individual candidate scoring. Our method bridges probabilistic and semantic perspectives, enabling efficient per-sample ranking.

3 Methodology

3.1 Problem Statement

Given a prompt x, we sample N independent generations $\{y_1, \dots, y_N\}$ from a language model using multinomial decoding. Each generation is mapped to a final answer $\{a_i\}_{i=1}^N$, optionally associated with generation likelihoods $\{p(y_i|x)\}_{i=1}^N$. Our goal is to identify the most reliable answer among the candidates. Formally, we consider the problem:

$$a^* = \arg\max_{a_i} \; Q(a_i|x), \tag{1}$$

where $Q(a_i|x)$ denote an unknown quality function reflecting the correctness or reliability of an answer, which must be estimated from a set of sampled generations.

3.2 Answer Selection via Geometric Consensus

To approximate the latent quality function $Q(a_i|x)$, we propose to model agreement among answers in a continuous semantic space. Formally, we embed each final answer into a vector space:

$$\mathbf{u}_i = \mathbf{E}(a_i), \tag{2}$$

where $\mathbf{E}$ is a sentence embedding model. Intuitively, in the context of best-of-N, sampled answers form a distribution of semantic interpretations of the same query, and a representative solution should reflect their overall agreement. The Fréchet mean [13] defines such a centroid as the point that minimizes the expected squared distance to all candidate embeddings in the semantic space. Given embeddings $\{\mathbf{u}_i\}_{i=1}^N \subset \mathbb{R}^d$ with a probability distribution $P{=}\{p_i\}_{i=1}^N$, the semantic center is defined as the Fréchet mean under squared Euclidean loss:

$$\mathbf{c}(P) = \arg\min_{\mathbf{z}} \sum_{i=1}^{N} p_i \|\mathbf{u}_i - \mathbf{z}\|_2^2, \tag{3}$$

This admits the closed-form solution:

$$\mathbf{c}(P) = \sum_{i=1}^{N} p_i \mathbf{u}_i, \tag{4}$$

(see Appendix A.1 for details).

We adopt the squared Euclidean loss for its simple closed-form solution. When the center is restricted to the discrete set $\{\mathbf{u}_i\}_{i=1}^N$, the problem reduces to selecting a representative sample:

$$\mathbf{c}_{\text{medoid}}(P) = \arg\min_{\mathbf{u}_j} \sum_{i=1}^{N} p_i \|\mathbf{u}_i - \mathbf{u}_j\|_2^2, \tag{5}$$

which corresponds to a weighted medoid. This can be viewed as a discrete counterpart of the Fréchet mean, where the solution is constrained to be one of the observed samples.

Different choices of P (Table 1) induce different notions of semantic consensus. In particular, the uniform distribution recovers the standard unweighted center, while frequency-based or likelihood-based distributions incorporate empirical or model-derived reliability signals. Conceptually, $\mathbf{c}(P)$ represents a consensus embedding of the generated answers, with semantically similar answers pulling the center toward their shared meaning and divergent or noisy samples contributing proportionally to their weights in P (see Figure 1).

3.3 Ranking via Radial Consensus Score

Given a semantic center $\mathbf{c}(P)$, we define a ranking function based on the radial dispersion of each candidate embedding relative to this center. For each candidate embedding $\mathbf{u}_i$, its **Radial Consensus Score** (RCS) is defined as:

$$\mathrm{RCS}_i(P) = \|\mathbf{u}_i - \mathbf{c}(P)\|_2. \tag{6}$$

RCS measures how semantically aligned each candidate is with the global consensus induced by the distribution P: smaller values indicate stronger agreement with the collective semantic structure, while larger values suggest divergence or potential noise. We adopt the ℓ_2 norm for its simplicity and consistency with the center formulation (Equation 3), as the choice of norm does not materially affect the relative ranking of candidates in practice.

The final prediction is selected as the candidate with minimum radial dispersion to the estimated consensus:

$$a^*(P) = \arg\min_{a_i} \mathrm{RCS}_i(P). \tag{7}$$

In discrete settings, the above objective admits an equivalent **weighted medoid** formulation:

$$a^*(P) = \arg\min_{a_i} \sum_{j} P(a_j)\, \|\mathbf{u}_i - \mathbf{u}_j\|_2^2. \tag{8}$$

This selects the candidate whose embedding minimizes the expected *squared* distance to others under P, consistent with the squared-Euclidean Fréchet mean objective (Equation 3). When P is uniform, this reduces to the standard medoid.

Table 1: Different instantiations of the distribution P in RCS, where f_i and $\Pr(y_i|\mathbf{x})$ denote the empirical frequency and model probability of candidate y_i, respectively. For the frequency-weighted variant, the index i runs over the set of *unique* answers, so that $\sum_j f_j = N$. For the cosine-similarity-weighted variant, $\cos(\mathbf{u}_i, \mathbf{u}_j)$ denotes the cosine similarity between L_2-normalized embeddings.

Variant	**Distribution** P	**Note**
Uniform	$p_i = \frac{1}{N}$	Black-box
Frequency-weighted	$p_i = \frac{f_i}{\sum_j f_j}$	Black-box
Probability-weighted	$p_i \propto \Pr(y_i|\mathbf{x})$	White-box (calibrated)
Cosine-weighted	$p_i \propto \frac{1}{N}\sum_{j=1}^{N}\cos(\mathbf{u}_i, \mathbf{u}_j)$	Black-box

Theoretical Analysis. To formally characterize the behavior of RCS, we model the answer embeddings as draws from a Gaussian noise distribution centered at the true semantic center–that is, $\mathbf{u}_i = \mathbf{c}^* + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. Under this generative assumption, we establish two propositions showing that RCS is strictly superior to majority voting, formulated as follows. The uniform, frequency, and probability-weighted variants admit direct proofs via the strong law of large numbers; the cosine-similarity-weighted variant satisfies the same asymptotic guarantee under an additional regularity condition (Appendix A.2).

Proposition 1 (**Asymptotic Superiority**). Under the Gaussian generative model, for *any* of the four distributions P (Table 1), as $N \to \infty$,

$$\mathbf{c}(P) \xrightarrow{a.s.} \mathbf{c}^*, \tag{9}$$

by the strong law of large numbers. Consequently,

$$\mathbb{P}(\text{RCS selects } a^*) \to 1, \quad \mathbb{P}(\text{voting selects } a^*) \leq \frac{1}{M}, \tag{10}$$

where a^* is the true best answer (corresponding to $\mathbf{c}^*$) and $M \geq 2$ is the number of distinct semantic clusters (typically $M \gg 1$ in open-ended generation). Thus RCS is asymptotically strictly superior to majority voting whenever lexical variance (due to paraphrasing) exceeds semantic variance.

Proof. See Appendix A.2. □

Proposition 2 (**Finite-N Gap with 2-Cluster Toy Model**). Consider the following two-cluster setting: (1) Cluster A (true high-quality): $k \geq N/2$ samples centered at $\mathbf{c}^* + \boldsymbol{\delta}$, $\|\boldsymbol{\delta}\|_2$ small; and (2) Cluster B (spurious): $N - k$ samples centered at $\mathbf{c}^* + \boldsymbol{\Delta}$, $\|\boldsymbol{\Delta}\|_2 \gg \|\boldsymbol{\delta}\|_2$.

Under the frequency-weighted distribution, the empirical center $\mathbf{c}(P)$ is geometrically biased toward Cluster A when $k \geq N/2$. Approximating each cluster's candidates at their expected locations, RCS selects Cluster A with probability

$$\mathbb{P}(\text{RCS correct}) = \Phi\left(\frac{(2k - N)\big(\|\boldsymbol{\Delta}\|_2 - \|\boldsymbol{\delta}\|_2\big)}{2\sigma\sqrt{N}}\right), \tag{11}$$

where Φ is the standard normal CDF. Under exact string matching with $m \geq 2$ distinct surface forms per semantic cluster, the most frequent correct form receives at most k/m out of N votes; a single random sample is correct with probability k/N.

Therefore,

$$\mathbb{P}(\text{RCS correct}) > \frac{k}{N} \tag{12}$$

whenever the semantic gap satisfies

$$\|\boldsymbol{\Delta}\|_2 - \|\boldsymbol{\delta}\|_2 > C\frac{\sigma}{\sqrt{N}}, \quad C > 0. \tag{13}$$

This property holds for the uniform and probability-weighted variants.

Remark. When $k < N/2$, the center $\mathbf{c}(P)$ is biased toward Cluster B and the sufficient condition above does not hold. Proposition 1 still guarantees $\mathbf{c}(P) \xrightarrow{a.s.} \mathbf{c}^*$ as $N \to \infty$, ensuring asymptotic correctness for all k.

Proof. See Appendix A.3. $\square$

Overall, the choice of P controls how consensus is formed: uniform weighting reflects pure geometric agreement, frequency/probability weighting emphasizes stable or high-confidence responses, and cosine-similarity weighting uses mutual embedding agreement—all naturally suppressing inconsistent outputs. RCS_{uni}, RCS_{freq}, and $\text{RCS}_{\text{cosine}}$ operate purely on model outputs and embeddings, making them applicable in black-box settings. While RCS is defined over candidate-level embeddings, one may alternatively consider aggregating representations over full generation trajectories. However, as shown in Section 4.3, such approaches suffer from representation collapse and lead to degraded performance, highlighting the importance of focusing on final-answer semantics. We provide the detailed pseudo-code for our method in Appendix Algorithm 1.

4 Experiment

4.1 Experiment Setup

Datasets. We use seven established benchmarks covering both short-form question-answering and long-form reasoning tasks with different answer formats: (1) Short-form QA: **SciQ** [47] and GPQA Diamond (**GPQA** [37]), (2) Long-form Math Reasoning: **Arithmetics** [6], **GSM8K** [7], and (3) Long-form QA multiple choice: MMLU Formal Logic (**Form.Log.** [18]), **MMLU-Pro** [45], Big-BenchHard (**BBH-Date**, **BBH-Navigate** [38]).

Models. We evaluate on four popular open-weight models from distinct families: Qwen2.5-3B [39], Llama3.2-3B, Llama3.1-8B [15], and Gemma2-9B [29].

Baselines. We compare our method against the following baselines, which require only single generation pass: (1) **NLL** [32], which selects answers with the highest likelihood based on negative log-likelihood; (2) **ANLL** [16], which uses average negative log-likelihood for length-normalized scoring; (3) **Self-Consistency (SC)** [44], selecting the most frequent answer; (4) **Self-Certainty (CE)** [21], which ranks answers by combining frequency and model confidence using Borda Voting; (5) **ModeX** [5], which applies spectral clustering over full reasoning trajectories and selects answers from the resulting cluster centroids; and (6) **SCW** [20], which embeds full reasoning trajectories, computes pairwise cosine similarities, groups scores by unique reasoning path, and selects the highest-scoring group. We report four RCS variants (Table 1): RCS_{uni}, RCS_{freq}, RCS_{prob} (us-

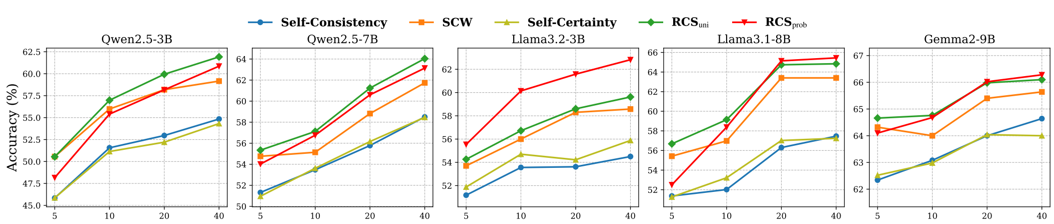

Figure 2: Average performance over five benchmarks for different numbers of sampling responses N. Each subplot corresponds to an LLM backbone, showing how accuracy changes with increasing N.

ing ANLL as a length-normalized probability proxy), and $\text{RCS}_{\text{cosine}}$ (per-sample cosine-similarity weights applied to final-answer embeddings, bypassing SCW's full-trajectory collapse and discrete grouping). We also report $\text{RCS}_{\text{medoid}}$ (uniform, unweighted) as a discrete baseline. All RCS variants use `all-MiniLM-L6-v2` [36]. We sample N=10 responses (also N=5, 20, 40) at T=1 via vLLM [24]; results are mean$\pm$std over three seeds. Additional details are in Appendix A.6.

Table 2: Best-of-N selection accuracy. **BBF**: black-box friendly. For non-RCS methods, **bold** = statistically significant improvement over best RCS (p<0.05, paired t-test). For RCS methods (shaded), **bold** = best RCS variant per column. Greedy averages use unrounded values. Additional results in Appendix Table 8.

Model	Method	BBF	SciQ	GPQA	Arithmetics	GSM8K	Form.Log.	Average
Qwen2.5-3B	Greedy	✓	59.3	20.9	89.0	65.0	34.9	54.3
	Oracle	✓	80.8±0.7	51.0±2.9	99.5±0.5	93.1±0.8	73.5±5.7	79.6
	NLL	✗	59.4±0.7	15.7±2.4	36.0±0.9	25.0±3.1	23.0±2.7	31.8
	ANLL	✗	59.2±0.5	15.9±2.2	36.0±1.3	24.9±3.3	22.8±3.0	31.8
	SC	✓	64.4±0.6	21.4±0.6	78.7±3.3	52.8±1.3	40.5±2.1	51.6
	CE	✗	64.0±0.5	16.8±1.7	80.3±4.3	53.3±2.8	41.3±0.8	51.1
	ModeX	✓	62.3±0.1	18.1±2.3	48.5±2.3	33.3±3.1	25.1±2.6	37.5
	SCW	✓	64.5±0.8	23.0±0.3	83.7±3.3	64.3±2.3	44.4±0.8	56.0
	RCS_{medoid}	✓	65.2±1.0	23.8±0.5	81.7±3.0	63.6±2.6	43.6±1.6	55.6
	RCS_{uni}	✓	**65.6±0.4**	23.1±2.2	**85.7±3.2**	**65.8±3.1**	**44.7±1.2**	**57.0**
	RCS_{freq}	✓	64.8±0.7	22.8±0.5	82.8±4.6	59.7±1.6	43.4±1.7	54.7
	RCS_{cosine}	✓	**65.6±0.6**	**24.2±1.8**	85.2±4.5	65.0±2.8	**44.7±1.2**	56.9
	RCS_{prob}	✗	65.5±0.6	22.5±2.4	84.2±4.0	63.4±1.8	43.9±1.2	55.9
Llama3.2-3B	Greedy	✓	55.6	23.0	96.0	79.7	34.9	58.3
	Oracle	✓	79.8±0.4	52.5±1.5	99.8±0.3	92.6±0.3	71.4±4.4	79.2
	NLL	✗	53.6±1.1	20.2±3.2	86.5±1.3	74.3±1.5	**33.3±2.1**	53.6
	ANLL	✗	53.6±0.5	19.7±0.5	88.5±2.6	73.9±2.4	**29.9±6.0**	53.1
	SC	✓	**58.8±1.1**	17.4±2.6	**97.5±1.3**	78.0±0.8	16.1±3.2	53.6
	CE	✗	**59.3±0.6**	20.9±2.2	**97.3±1.3**	78.5±0.8	17.5±2.7	54.7
	ModeX	✓	53.2±0.6	18.1±2.1	70.0±2.3	62.6±2.1	16.9±0.5	44.2
	SCW	✓	59.5±1.0	25.4±0.9	94.0±1.3	80.5±1.5	20.6±2.4	56.0
	RCS_{medoid}	✓	**60.0±1.2**	25.6±0.8	**98.3±1.0**	80.5±1.5	19.6±3.7	56.8
	RCS_{uni}	✓	59.5±0.8	25.0±0.8	98.2±1.0	80.5±1.5	20.4±3.2	56.7
	RCS_{freq}	✓	59.8±1.0	25.4±0.9	98.2±1.0	79.4±1.3	18.0±3.7	56.2
	RCS_{cosine}	✓	59.8±1.2	25.2±1.2	98.0±0.9	81.1±1.6	20.6±3.6	56.9
	RCS_{prob}	✗	59.6±0.8	**26.2±1.3**	97.5±0.8	**84.1±0.8**	**33.1±3.2**	**60.2**
Llama3.1-8B	Greedy	✓	63.1	25.5	89.5	88.3	34.9	60.8
	Oracle	✓	85.1±0.4	56.3±1.3	99.3±0.3	94.6±0.8	74.1±1.2	81.9
	NLL	✗	61.0±0.5	20.4±0.8	73.3±2.8	69.0±0.5	33.9±2.8	51.5
	ANLL	✗	61.1±0.5	18.7±0.9	73.5±4.1	68.7±1.6	33.1±2.4	51.0
	SC	✓	**67.3±0.7**	16.8±1.8	78.3±3.0	66.0±2.3	31.7±2.1	52.0
	CE	✗	66.8±0.9	18.5±1.1	81.0±3.8	67.5±1.3	32.3±1.7	53.2
	ModeX	✓	63.1±1.2	17.8±1.1	55.3±2.5	48.4±3.9	25.1±2.6	41.9
	SCW	✓	67.7±1.0	26.8±0.8	82.2±2.8	71.4±2.9	36.8±1.7	57.0
	RCS_{medoid}	✓	68.0±0.9	26.4±0.9	86.5±1.5	71.4±2.8	37.8±0.9	58.0
	RCS_{uni}	✓	**68.1±0.7**	26.9±1.0	**88.5±0.5**	72.8±3.0	**39.4±0.9**	**59.1**
	RCS_{freq}	✓	67.6±1.1	26.9±1.0	83.7±2.3	68.9±1.9	34.9±0.8	56.4
	RCS_{cosine}	✓	**68.1±0.7**	27.3±0.8	87.7±1.2	72.8±2.3	**39.4±0.9**	**59.1**
	RCS_{prob}	✗	67.9±1.1	**27.5±0.9**	85.2±0.6	**75.6±1.0**	36.8±2.6	58.6

4.2 Main Results

RCS Consistently Outperforms Baselines. We report selection accuracy in Table 2. RCS variants consistently achieve the best performance across all settings. RCS_{uni} and RCS_{prob} rank first overall, outperforming the best baseline (CE) by 2–6%. RCS_{cosine} consistently ranks second among RCS variants (avg. 56.9–59.1), outperforming SCW (avg. 56.0–57.0) despite using the same pairwise cosine signals—demonstrating that the RCS Fréchet-mean framework extracts more value from the same information by bypassing the discrete grouping step. RCS_{freq} outperforms SC by avoiding hard tie-breaking through soft semantic weighting. Among baselines, CE is the strongest competitor. ModeX and SCW both underperform due to full-trajectory embedding collapse and, in SCW's case, the additional discrete-voting step that misses minority correct answers (Section 4.3). Overall, results confirm that a unified geometric framework with flexible weighting distributions is both more expressive and more robust than fixed surface-form or trajectory-level aggregation.

RCS Scales Effectively with Increasing Sampling Budget. Figure 2 shows performance as the number of sampled responses N increases with only top methods are shown. We highlight RCS_{uni} and RCS_{prob}, the best-performing variants from Table 2 (full results in Appendix A.7). RCS consistently outperforms all baselines at larger N (e.g., N=20, 40), with gains up to +7%. As N increases, correct answers may remain in the minority; RCS recovers them by anchoring to the semantic center rather than to frequency, maintaining robustness even when correct outputs are sparse.

4.3 Ablation Study

To reduce computational overhead, experiments are conducted using two representative models: Qwen2.5-3B and Llama3.2-3B.

Table 3: Performance on Arithmetics and Form.Log. with Multi-agent Debate. Best values are bolded. **R=0,1,2** indicate debate rounds, while **Avg.** denotes the average over all rounds. $\text{RCS}_{\text{base}} \equiv \text{RCS}_{\text{uni}}$ (uniform weighting).

Model	Method	N	Arithmetics				Form.Log.			
			R=0	**R=1**	**R=2**	**Avg.**	**R=0**	**R=1**	**R=2**	**Avg.**
Qwen2.5-3B	SC	10	94.5±4.0	60.4±1.6	34.6±5.8	63.1	44.8±2.5	35.3±2.9	27.6±5.4	35.9
	RCS_{base}	10	94.3±3.7	59.5±4.1	34.0±11.0	62.6	**46.0±2.3**	35.8±5.3	**28.8±6.0**	**36.9**
	RCS_{freq}	10	**95.2±3.6**	**61.4±1.2**	**34.7±5.4**	**63.8**	44.8±2.8	**36.1±3.3**	28.7±5.5	36.5
	SC	20	96.5±0.5	24.8±3.0	34.0±3.3	51.8	42.3±0.5	40.5±1.6	35.7±1.6	39.5
	RCS_{base}	20	**97.7±1.0**	**27.3±3.2**	**34.5±5.1**	**53.2**	**46.6±0.9**	**45.2±1.4**	**37.8±0.9**	**43.2**
	RCS_{freq}	20	97.2±0.3	26.0±2.2	34.3±4.0	52.5	42.6±0.5	42.1±0.8	35.7±1.6	40.1
Llama3.2-3B	SC	10	80.7±2.0	85.6±10.7	75.8±5.9	80.7	31.3±2.9	28.4±7.8	29.5±7.7	29.7
	RCS_{base}	10	79.1±4.3	**85.8±9.1**	**76.2±4.6**	80.4	**33.2±0.5**	**28.7±8.0**	**29.9±6.4**	**30.6**
	RCS_{freq}	10	**81.1±12.7**	85.6±10.0	**76.2±5.1**	**81.0**	32.1±2.9	28.6±8.3	29.5±16.6	30.1
	SC	20	99.0±0.5	72.2±4.1	41.5±2.2	70.9	13.2±0.5	3.2±0.8	3.7±1.7	6.7
	RCS_{base}	20	99.0±1.0	**76.5±4.3**	**43.3±1.5**	**72.9**	**21.4±2.4**	**9.0±1.2**	**7.1±2.4**	**12.5**
	RCS_{freq}	20	**99.2±0.8**	73.8±3.8	42.5±2.6	71.8	15.6±1.7	4.2±0.5	3.7±0.9	7.8

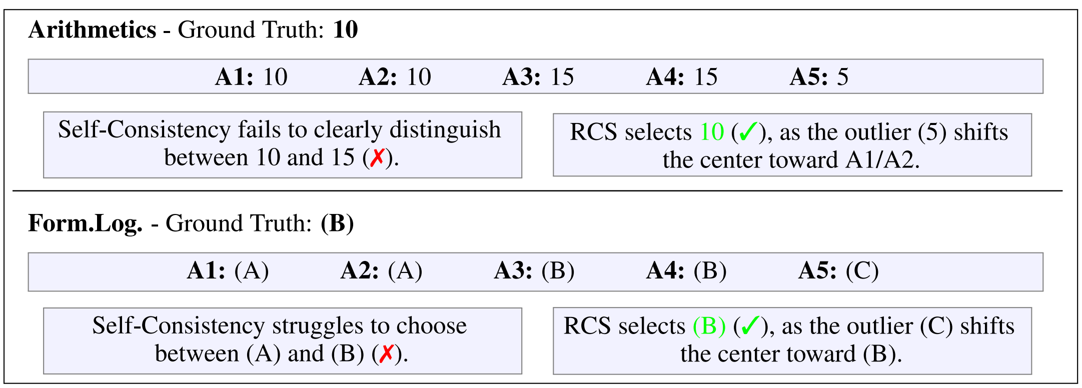

Figure 3: Qualitative examples showing RCS recovers correct answers more reliably than SC.

RCS Enhances Multi-Agent Debate. We further evaluate RCS as a drop-in replacement for majority voting in multi-agent debate [11, 6] under a fully black-box setting with N=10, 20 agents and R=2 rounds. We report RCS_{base} and RCS_{freq} (Table 3), which are computationally efficient and do not require access to model internals. Increasing the number of agents to N=20 consistently degrades performance across debate rounds, particularly on Llama3.2-3B, suggesting that naive scaling introduces additional variability that destabilizes reasoning [33]. In contrast, RCS improves performance by 1–2% on average, with larger gains at N=20, highlighting its robustness when aggregation becomes more challenging.

RCS Improves Under Fully Black-box Setting. We further evaluate a fully black-box setting using Cohere (*command-a-03-2025*) on GPQA, MMLU-Pro, and BigBenchHard (Table 4a). RCS outperforms traditional Self-Consistency, with RCSbase achieving the best performance, yielding +1% average improvement. This highlights the advantage of RCS in capturing consistency beyond simple frequency-based selection.

Table 4: Performance under (a) black-box setting (N=10) and (b) complexity comparison.

(a)

Method	GPQA	MMLU-Pro	BBH-Date	BBH-Navigate	Avg.
SC	13.1	47.6	76.8	39.2	44.2
$\text{RCS}_{\text{medoid}}$	**14.6**	**48.6**	76.4	39.2	44.7
RCS_{base}	14.1	**48.6**	**78.0**	40.2	**45.2**
RCS_{freq}	13.6	47.6	77.6	**41.2**	45.0

(b)

Method	Complexity
NLL, ANLL	$\mathcal{O}(NL)$
SC	$\mathcal{O}(N)$
CE	$\mathcal{O}(NL)$
ModeX	$\mathcal{O}(N^2L + N^3)$
SCW	$\mathcal{O}(N^2d)$
$\text{RCS}_{\text{medoid}}$	$\mathcal{O}(N^2d)$
$\text{RCS}_{\text{uni/freq}}$	$\mathcal{O}(Nd)$
RCS_{prob}	$\mathcal{O}(N(d+L))$
$\text{RCS}_{\text{cosine}}$	$\mathcal{O}(N^2d)$

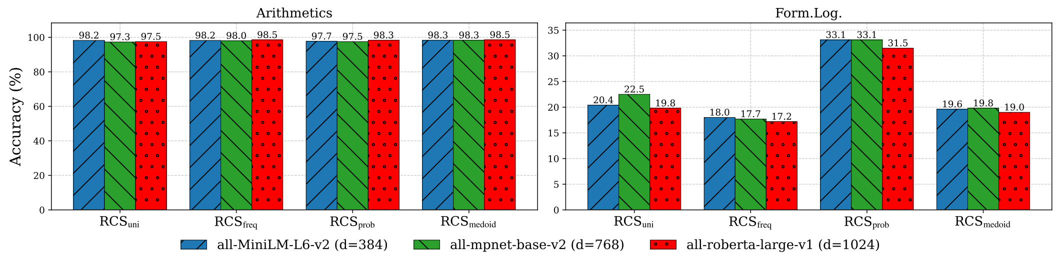

Figure 4: Effect of encoders on Llama3.2-3B. Results for Qwen2.5-3B are in Appendix Figure 6.

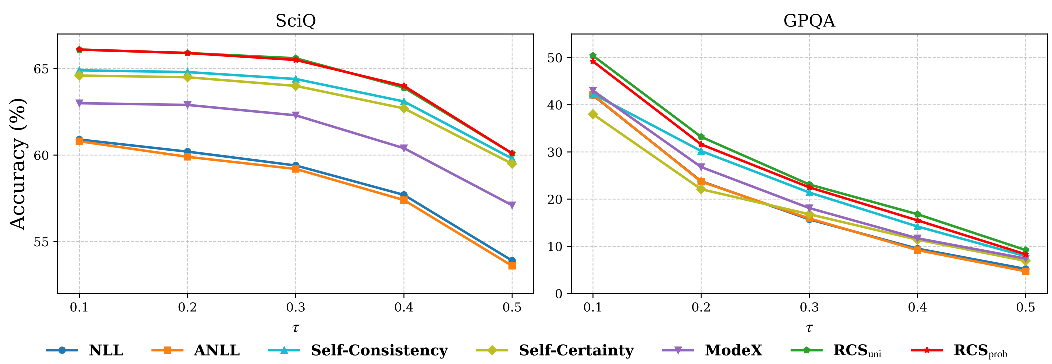

Figure 5: Effect of varying τ using Qwen2.5-3B. Results for Llama3.2-3B are in Appendix Figure 7.

Complexity Analysis. We report computational complexity in Table 4b, where d, L, and $|V|$ denote embedding dimension, sequence length, and vocabulary size. NLL and ANLL scale as $\mathcal{O}(NL)$ due to token-level scoring, while SC is $\mathcal{O}(N)$ since it only performs frequency counting over final answers. More expressive methods such as CE and ModeX capture richer semantic or structural dependencies but incur higher costs due to full token-level distributions or pairwise interactions. In contrast, RCS variants (e.g., RCS_{uni}) avoid pairwise comparisons by estimating a semantic center and dispersion in $\mathcal{O}(Nd)$, achieving a favorable trade-off between efficiency and semantic expressiveness.

Qualitative Example. We present two qualitative examples from Arithmetics and Form.Log. (Figure 3), using N=5 samples for Self-Consistency and RCS_{uni} for simplicity. Self-Consistency fails when majority responses are misleading, whereas RCS_{uni} correctly recovers the answer by leveraging informative minority samples. This highlights the robustness of RCS in aggregating diverse outputs under noisy sampling.

Analysis Of Design Choices. We analyze the impact of several design choices in RCS. **(1) Effect Of Embedding Models:** In Figure 4, we compare three sentence transformers of increasing capacity. Results show minimal variation across models. This stability is especially evident on Arithmetics, where answers are numeric. On Form.Log., we observe a slight drop with `all-roberta-large-v1`, particularly for the probability-based variant, suggesting that higher embedding capacity does not necessarily improve semantic alignment in this setting. **(2) Effect Of Correctness Threshold:** Figure 5 reports performance on SciQ and GPQA under varying correctness thresholds. We focus on RCS_{uni} and RCS_{prob}, which consistently achieve the best results as the ROUGE-F1 threshold increases from 0.1 to 0.5, particularly within the common range of 0.2–0.4. **(3) Effect of Full-Trajectory Embeddings:** A natural extension of RCS is to use full generation trajectories instead of final-answer embeddings. However, this leads to an *embedding collapse* effect, where candidates become less distinguishable due to shared prefixes and redundant reasoning. We compare this variant with final-answer embeddings (Appendix Figures 8 and 9) and observe consistent performance drops. This suggests that intermediate tokens introduce noise and weaken discriminative signals, while final answers provide a more compact and task-relevant representation. This collapse issue also affects methods that operate on full trajectories: ModeX applies spectral clustering over complete reasoning traces, while SCW [20] embeds full reasoning paths and groups pairwise cosine-similarity scores by unique trajectory. Although SCW's similarity-based aggregation provides an implicit outlier-down-weighting effect, shared prefixes still cause representations to collapse, and the discrete grouping-by-trajectory step can miss minority correct answers. RCS avoids both issues by restricting embeddings to the extracted final answer and ranking all candidates continuously via the weighted Fréchet mean.

5 Conclusion

We introduce Radial Consensus Score (RCS), a geometric framework for best-of-N answer selection in LLMs. RCS ranks candidates by their distance to a weighted Fréchet mean of answer embeddings, with the weighting scheme—uniform, frequency, probability, or cosine-similarity—determining how consensus is defined. Compared to Semantic Consensus Weighting (SCW), which applies the same cosine signal via a discrete grouping step, $\text{RCS}_{\text{cosine}}$ uses it continuously as Fréchet-mean weights, recovering higher-quality minority answers that discrete voting misses. RCS variants consistently outperform strong baselines across diverse benchmarks, with gains growing at higher sampling budgets, while remaining black-box compatible and applicable as a drop-in replacement for majority voting in multi-agent debate.

References

[1] Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe, and Haifeng Xu. Beyond majority voting: Llm aggregation by leveraging higher-order information. *arXiv preprint arXiv:2510.01499*, 2025.

[2] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

[3] Jiahao Chen et al. Latent self-consistency for reliable majority-set selection in short- and long-answer reasoning. *arXiv preprint*, 2024.

[4] Xinyun Chen, Renat Aksitov, Uri Alon, Jie Ren, Kefan Xiao, Pengcheng Yin, Sushant Prakash, Charles Sutton, Xuezhi Wang, and Denny Zhou. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*, 2023.

[5] Hyeong Kyu Choi and Sharon Li. Modex: Evaluator-free best-of-n selection for open-ended generation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026.

[6] Hyeong Kyu Choi, Xiaojin Zhu, and Sharon Li. Debate or vote: Which yields better decisions in multi-agent large language models? *arXiv preprint arXiv:2508.17536*, 2025.

[7] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[8] Paula Cordero-Encinar and Andrew B Duncan. Certified self-consistency: Statistical guarantees and test-time training for reliable reasoning in llms. *arXiv preprint arXiv:2510.17472*, 2025.

[9] Hiroyuki Deguchi et al. Centroid-based efficient minimum bayes risk decoding. *arXiv preprint arXiv:2402.10084*, 2024.

[10] Qiwei Di, Kaixuan Ji, Xuheng Li, Heyang Zhao, and Quanquan Gu. Best-of-majority: Minimax-optimal strategy for pass@ k inference scaling. *arXiv preprint arXiv:2510.03199*, 2025.

[11] Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Forty-first International Conference on Machine Learning*, 2023.

[12] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

[13] P.T. Fletcher, Conglin Lu, S.M. Pizer, and Sarang Joshi. Principal geodesic analysis for the study of nonlinear statistics of shape. *IEEE Transactions on Medical Imaging*, 23(8):995–1005, 2004.

[14] Yichao Fu, Xuewei Wang, Yuandong Tian, and Jiawei Zhao. Deep think with confidence. *The Fourteenth International Conference on Learning Representations*, 2026.

[15] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[16] Nuno M Guerreiro, Elena Voita, and André FT Martins. Looking for a needle in a haystack: A comprehensive study of hallucinations in neural machine translation. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1059–1075, 2023.

[17] Lin Gui, Cristina Gârbacea, and Victor Veitch. Bonbon alignment for large language models and the sweetness of best-of-n sampling. *Advances in Neural Information Processing Systems*, 37:2851–2885, 2024.

[18] Dan Hendrycks, Collin Burns, Steven Basart, Andrew Critch, Jerry Li, Dawn Song, and Jacob Steinhardt. Aligning ai with shared human values. *arXiv preprint arXiv:2008.02275*, 2020.

[19] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55, 2025.

[20] Harsh Jain et al. Semantic self-consistency: Enhancing language model reasoning via semantic weighting. *arXiv preprint arXiv:2401.06640*, 2024.

[21] Zhewei Kang, Xuandong Zhao, and Dawn Song. Scalable best-of-n selection for large language models via self-certainty. *arXiv preprint arXiv:2502.18581*, 2025.

[22] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *The Eleventh International Conference on Learning Representations*, 2023.

[23] Lucie Kunitomo-Jacquin, Edison Marrese-Taylor, Ken Fukuda, and Masahiro Hamasaki. Evidential semantic entropy for LLM uncertainty quantification. In Vera Demberg, Kentaro Inui, and Lluís Marquez, editors, *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7107–7122, Rabat, Morocco, March 2026. Association for Computational Linguistics.

[24] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*, 2023.

[25] Xiaomin Li, Zhou Yu, Ziji Zhang, Yingying Zhuang, Swair Shah, Narayanan Sadagopan, and Anurag Beniwal. Semantic volume: Quantifying and detecting both external and internal uncertainty in LLMs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 31751–31759, 2026.

[26] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *The twelfth international conference on learning representations*, 2023.

[27] Zhen Lin, Shubhendu Trivedi, and Jimeng Sun. Generating with confidence: Uncertainty quantification for black-box large language models. *arXiv preprint arXiv:2305.19187*, 2023.

[28] Potsawee Manakul, Adian Liusie, and Mark Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017, 2023.

[29] Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. Gemma: Open models based on Gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.

[30] Niels Mündler et al. Semantic density: Uncertainty quantification for large language models through confidence measurement in semantic space. *arXiv preprint arXiv:2405.13845*, 2024.

[31] Manh Nguyen, Sunil Gupta, and Hung Le. Distance is all you need: Radial dispersion for uncertainty estimation in large language models. *arXiv preprint arXiv:2512.04351*, 2025.

[32] Manh Nguyen, Sunil Gupta, and Hung Le. Probabilities are all you need: A probability-only approach to uncertainty estimation in large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 32546–32554, 2026.

[33] Manh Nguyen, Anh Nguyen, Dung Nguyen, Svetha Venkatesh, and Hung Le. Hear both sides: Efficient multi-agent debate via diversity-aware message retention. *arXiv preprint arXiv:2603.20640*, 2026.

[34] Alexander Nikitin, Jannik Kossen, Yarin Gal, and Pekka Marttinen. Kernel language entropy: Fine-grained uncertainty quantification for llms from semantic similarities. *Advances in Neural Information Processing Systems*, 37:8901–8929, 2024.

[35] Xin Qiu and Risto Miikkulainen. Semantic density: Uncertainty quantification for large language models through confidence measurement in semantic space. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.

[36] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019.

[37] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. GPQA: A graduate-level Google-proof Q&A benchmark. In *First Conference on Language Modeling*, 2024.

[38] Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc Le, Ed Chi, Denny Zhou, et al. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051, 2023.

[39] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.

[40] Guangya Wan, Yuqi Wu, Jie Chen, and Sheng Li. Reasoning aware self-consistency: Leveraging reasoning paths for efficient llm sampling. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3613–3635, 2025.

[41] Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9426–9439, Bangkok, Thailand, August 2024. Association for Computational Linguistics.

[42] Weiqin Wang, Yile Wang, and Hui Huang. Ranked voting based self-consistency of large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14410–14426, 2025.

[43] Xinglin Wang, Yiwei Li, Shaoxiong Feng, Peiwen Yuan, Boyuan Pan, Heda Wang, Yao Hu, and Kan Li. Integrate the essence and eliminate the dross: Fine-grained self-consistency for free-form language generation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11782–11794, 2024.

[44] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.

[45] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37:95266–95290, 2024.

[46] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

[47] Johannes Welbl, Nelson F Liu, and Matt Gardner. Crowdsourcing multiple choice science questions. *arXiv preprint arXiv:1707.06209*, 2017.

[48] Tunyu Zhang, Haizhou Shi, Yibin Wang, Hengyi Wang, Xiaoxiao He, Zhuowei Li, Haoxian Chen, Ligong Han, Kai Xu, Huan Zhang, et al. Tokur: Token-level uncertainty estimation for large language model reasoning. *The Fourteenth International Conference on Learning Representations*, 2026.

A Appendix

Algorithm 1 Best-of-N Selection with Radial Consensus Score (RCS)

Require: Answers $\{a_i\}_{i=1}^N$, distribution P, embedding model $\mathbf{E}$, mode $\in \{\text{continuous}, \text{discrete}\}$
Ensure: Selected index i^*

```
Embeddings: u_i = E(a_i)    for i = 1, ..., N
if mode = continuous then
    c ← Σ_{i=1}^N p_i u_i
else
    c ← arg min_{u_j} Σ_{i=1}^N p_i ||u_i − u_j||_2^2
end if
i* ← arg min_i ||u_i − c||_2
return i*
```

A.1 Proof of Geometric Center Estimation

Expanding the objective, we have:

$$\sum_{i=1}^{N} p_i \|\mathbf{u}_i - \mathbf{z}\|_2^2 = \sum_{i=1}^{N} p_i \left(\|\mathbf{u}_i\|_2^2 - 2\mathbf{u}_i^\top \mathbf{z} + \|\mathbf{z}\|_2^2\right) \tag{14}$$

$$= \sum_{i=1}^{N} p_i \|\mathbf{u}_i\|_2^2 - 2\mathbf{z}^\top \sum_{i=1}^{N} p_i \mathbf{u}_i + \|\mathbf{z}\|_2^2. \tag{15}$$

Taking derivative with respect to $\mathbf{z}$ and setting it to zero:

$$-2\sum_{i=1}^{N} p_i \mathbf{u}_i + 2\mathbf{z} = 0, \tag{16}$$

which yields:

$$\mathbf{z} = \sum_{i=1}^{N} p_i \mathbf{u}_i. \tag{17}$$

Since the objective is strictly convex in $\mathbf{z}$, this solution is unique.

A.2 Proof of Proposition 1 (Asymptotic Superiority)

Assumption of Gaussian Distribution. We model the embedding space $\mathbb{R}^d$ ($d \gg 1$) where the *embeddings* of the generated answers are given by

$$\mathbf{u}_i = \mathbf{c}^* + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad i.i.d., \tag{18}$$

with $\mathbf{c}^*$ the unknown true semantic center (embedding of the highest-quality answer) and $\boldsymbol{\varepsilon}_i$ capturing semantic noise such as paraphrasing or minor hallucinations. The discrete answers a_i themselves are produced by the LLM; the Gaussian assumption applies directly to their embeddings (the space relevant for RCS). Answer quality is inversely related to noise magnitude:

$$Q_i = Q(a_i|x) \propto -\|\boldsymbol{\varepsilon}_i\|_2. \tag{19}$$

The distribution $P{=}\{p_i\}_{i=1}^N$ can be uniform, frequency-weighted, or probability-weighted, as defined in Table 1. For the probability-weighted variant, we define

$$p_i = \frac{w(\boldsymbol{\varepsilon}_i)}{\sum_{j=1}^{N} w(\boldsymbol{\varepsilon}_j)}, \quad \text{where} \quad w(x) = \exp(-\lambda\|x\|_2^2),\ \lambda > 0. \tag{20}$$

This case arises naturally when the LLM is confidence-calibrated: its assigned probability decreases exponentially with semantic deviation.

Recall that the semantic center is the weighted average

$$\mathbf{c}(P) = \sum_{i=1}^{N} p_i \mathbf{u}_i = \mathbf{c}^* + \sum_{i=1}^{N} p_i \boldsymbol{\varepsilon}_i, \tag{21}$$

where the weights p_i satisfy $\sum_i p_i = 1$.

Under the Gaussian generative model $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ i.i.d., we have $\mathbb{E}[\mathbf{c}(P)] = \mathbf{c}^*$ for *any* of the three distributions P (uniform, frequency-weighted, or probability-weighted). Moreover,

$$\mathrm{Var}(\mathbf{c}(P)) = \sigma^2 \Big(\sum_{i=1}^{N} p_i^2 \Big) \mathbf{I}. \tag{22}$$

For uniform and frequency-weighted cases, the weights p_i are either deterministic or converge to a deterministic limit. By the weighted strong law of large numbers (SLLN) for i.i.d. random variables with $\max_i p_i \to 0$,

$$\sum_{i=1}^{N} p_i \boldsymbol{\varepsilon}_i \xrightarrow{a.s.} \mathbf{0}. \tag{23}$$

Hence $\mathbf{c}(P) \xrightarrow{a.s.} \mathbf{c}^*$.

For the probability-weighted variant, we rewrite the weighted sum as a self-normalized average:

$$\sum_{i=1}^{N} p_i \boldsymbol{\varepsilon}_i = \frac{\frac{1}{N} \sum_{i=1}^{N} w(\boldsymbol{\varepsilon}_i) \boldsymbol{\varepsilon}_i}{\frac{1}{N} \sum_{i=1}^{N} w(\boldsymbol{\varepsilon}_i)}. \tag{24}$$

By the strong law of large numbers, since both $w(\boldsymbol{\varepsilon}_i)\boldsymbol{\varepsilon}_i$ and $w(\boldsymbol{\varepsilon}_i)$ are i.i.d. with finite expectations,

$$\frac{1}{N} \sum_{i=1}^{N} w(\boldsymbol{\varepsilon}_i) \boldsymbol{\varepsilon}_i \xrightarrow{a.s.} \mathbb{E}[w(\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}], \quad \frac{1}{N} \sum_{i=1}^{N} w(\boldsymbol{\varepsilon}_i) \xrightarrow{a.s.} \mathbb{E}[w(\boldsymbol{\varepsilon})]. \tag{25}$$

By symmetry of the isotropic Gaussian distribution,

$$\mathbb{E}[w(\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}] = \mathbf{0}, \tag{26}$$

and $\mathbb{E}[w(\boldsymbol{\varepsilon})] > 0$, hence

$$\sum_{i=1}^{N} p_i \boldsymbol{\varepsilon}_i \xrightarrow{a.s.} \mathbf{0}. \tag{27}$$

Thus $\mathbf{c}(P) \xrightarrow{a.s.} \mathbf{c}^*$ also holds in this case.

The RCS selection rule is

$$\hat{i} = \arg\min_i \|\mathbf{u}_i - \mathbf{c}(P)\|_2. \tag{28}$$

As $\mathbf{c}(P) \to \mathbf{c}^*$ almost surely, we have

$$\hat{i} \to \arg\min_i \|\mathbf{u}_i - \mathbf{c}^*\|_2, \tag{29}$$

i.e., the sample with the smallest noise norm $\|\boldsymbol{\varepsilon}_i\|_2$. Because the minimum of N i.i.d. Gaussian norms concentrates near zero as $N \to \infty$,

$$\mathbb{P}(\text{RCS selects the true best sample}) \to 1. \tag{30}$$

In contrast, majority voting operates on exact string matches. Under paraphrasing, a single semantic cluster splits into $m \geq 2$ distinct surface forms, so the probability that any one form receives the strict majority is at most $1/m$. Hence

$$\mathbb{P}(\text{voting selects the true best answer}) \leq \frac{1}{m}. \tag{31}$$

This completes the proof for all three weight distributions. $\square$

A.3 Proof of Proposition 2 (Finite-N Gap with 2-Cluster Toy Model)

We consider the two-cluster generative model

$$\mathbf{u}_i \sim \begin{cases} \mathcal{N}(\mathbf{c}^* + \boldsymbol{\delta}, \sigma^2\mathbf{I}), & i \in A,\ |A| = k, \\ \mathcal{N}(\mathbf{c}^* + \boldsymbol{\Delta}, \sigma^2\mathbf{I}), & i \in B,\ |B| = N-k, \end{cases} \tag{32}$$

with $\|\boldsymbol{\delta}\|_2 \ll \|\boldsymbol{\Delta}\|_2$ (Cluster A = true high-quality; Cluster B = spurious), and assume $k \geq N/2$.

We first prove the result for the **frequency-weighted** distribution P. The semantic center is

$$\mathbf{c}(P) = \mathbf{c}^* + \tfrac{k}{N}\boldsymbol{\delta} + \tfrac{N-k}{N}\boldsymbol{\Delta} + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}\Big(\mathbf{0}, \tfrac{\sigma^2}{N}\mathbf{I}\Big). \tag{33}$$

Let $D := \|\boldsymbol{\Delta} - \boldsymbol{\delta}\|_2 \approx \|\boldsymbol{\Delta}\|_2 - \|\boldsymbol{\delta}\|_2$ and project onto $\mathbf{v} = (\boldsymbol{\Delta} - \boldsymbol{\delta})/D$. Define $\mu_A = \langle \mathbf{c}^* + \boldsymbol{\delta}, \mathbf{v}\rangle$. The projected coordinates are:

$$X_A \sim \mathcal{N}(\mu_A, \sigma^2), \quad X_B \sim \mathcal{N}(\mu_A + D, \sigma^2), \quad X_c \sim \mathcal{N}\Big(\mu_A + \tfrac{N-k}{N}D,\ \tfrac{\sigma^2}{N}\Big). \tag{34}$$

Since $k \geq N/2$, the center satisfies $|X_c^{\mathrm{mean}} - \mu_A| = \frac{N-k}{N}D \leq \frac{D}{2} \leq \frac{k}{N}D = |(\mu_A + D) - X_c^{\mathrm{mean}}|$, i.e. $\mathbf{c}(P)$ is geometrically closer to Cluster A than to Cluster B.

Treating each cluster's candidates at their expected locations ($X_A \approx \mu_A$, $X_B \approx \mu_A + D$), RCS selects Cluster A if and only if the center X_c falls below the midpoint $\mu_A + D/2$:

$$\mathbb{P}(\text{RCS selects A}) \approx \mathbb{P}\big(X_c < \mu_A + \tfrac{D}{2}\big) = \mathbb{P}\big(\eta_c < \tfrac{N-k}{N}D - \tfrac{D}{2}\big)^{-1} \ = \ \Phi\left(\frac{(2k-N)\,D}{2\sigma\sqrt{N}}\right), \tag{35}$$

where $\eta_c \sim \mathcal{N}(0, \sigma^2/N)$ and Φ is the standard normal CDF.

Comparison with voting. Under exact string matching with $m \geq 2$ distinct surface forms per cluster, the most frequent correct form receives at most k/m votes; a single uniformly random draw is correct with probability k/N. RCS exceeds this baseline whenever

$$\|\boldsymbol{\Delta}\|_2 - \|\boldsymbol{\delta}\|_2 > C\frac{\sigma}{\sqrt{N}}, \qquad C > 0. \tag{36}$$

Generalization to the other variants. For the **uniform** distribution the center computation is identical to frequency-weighted when each generated string is unique, so the bound carries over. For the **probability-weighted** distribution, $p_i \propto \exp(-\lambda\|\boldsymbol{\varepsilon}_i\|_2^2)$ assigns higher weight to Cluster A samples (which have smaller noise); the effective weight of A increases and $\mathrm{Var}(\mathbf{c}(P))$ shrinks by $O(1 + \lambda\sigma^2)$, so the success probability is at least as high as in the frequency-weighted case.

Remark on $k < N/2$. When $k < N/2$, the center $\mathbf{c}(P)$ is biased toward Cluster B ($|X_c^{\mathrm{mean}} - \mu_A| > D/2$) and the sufficient condition above does not hold. Proposition 1 still guarantees $\mathbf{c}(P) \xrightarrow{a.s.} \mathbf{c}^*$, ensuring asymptotic correctness for all k. $\square$

A.4 Limitations

While efficient, RCS relies on embedding quality and assumes that correct answers cluster around a semantic centroid. This assumption may be less effective in tasks with highly diverse or multimodal valid answers. Improving embedding robustness and better handling such diversity are promising directions for future work.

For multiple-choice tasks where the system prompt instructs models to output isolated option letters (e.g., `(A)`), continuous sentence embeddings of single characters do not inherently encode the semantic proximity between answer choices. In such settings, the geometric structure exploited by RCS may be less informative, though empirical results show consistent improvements even on these tasks (MMLU-Pro, BBH-Date, BBH-Navigate in Table 4).

The Gaussian generative model underlying Propositions 1 and 2 assumes isotropic noise and well-separated clusters. Real sentence embeddings often exhibit anisotropy (e.g., occupying a narrow cone), which may weaken the finite-N guarantee. The asymptotic result (Proposition 1) requires only unbiasedness and is more robust to such deviations.

A.5 Societal Impacts

RCS can positively impact applications by improving the reliability of best-of-N selection in LLMs, leading to more consistent and accurate outputs. However, its effectiveness depends on the quality of embedding representations, which may inherit or amplify biases from the underlying model. Therefore, caution is needed when applying it in high-stakes or safety-critical settings.

A.6 Implementation Details

Evaluation Protocol. For all methods, we measure accuracy (%) between the selected answer and the ground-truth. A generation is deemed correct via exact match on long-form reasoning tasks or ROUGE-L F1 > 0.3 on short-form QA tasks (SciQ, GPQA), exactly as in prior work [22, 32]. We include all model outputs, including empty or degenerate responses, to reflect real-world noisy generation settings.

We use 5-shot prompting [2] for short-form QA and Chain-of-Thought prompting [46] for long-form tasks. For CE, we follow the original setup and set p=0.3. We summarize the evaluation benchmarks, including the number of evaluation samples and representative examples, in Table 7. For MMLU-Pro, we sample up to 10 questions per category (105 samples across 14 categories; some categories contain fewer than 10 eligible questions, yielding an average of 7.5 per category) to ensure broad coverage; for BigBenchHard, we consider two challenging tasks: Date Understanding (**BBH-Date**) and Navigate (**BBH-Navigate**). Note that BBH-Navigate answers are binary (Yes/No); the system prompt requests a choice letter `(A)`, so parsed answers are mapped to the corresponding Yes/No option before evaluation. We observe no differences between breaking ties randomly and preserving the original answer order. For determinism and reproducibility, we adopt the latter. For the multi-agent debate setting, we follow prior implementations [6, 33]. Sampling hyperparameters and generation prompts are provided in Tables 5 and Tables 6, respectively. All experiments are conducted with three different seeds (42, 44, and 46) on a single H100-80GB GPU. Total compute across all models and benchmarks is approximately 120 GPU-hours; a single model evaluation (one benchmark, one model, N=10) completes in roughly 2–5 hours depending on model size.*

*Greedy averages in Tables 2 and 8 are computed from unrounded per-benchmark values; displayed per-column values are independently rounded to one decimal, which may produce a slight discrepancy versus the arithmetic mean of the displayed values.

Table 5: Sampling hyperparameters used for generation.

Parameter	**Value**
Temperature	1
Max new tokens (short-form QA)	32
Max new tokens (long-form tasks)	512

Table 6: System prompts used for generation across tasks.

Tasks	**Prompt**
SciQ, GPQA	This is a bot that correctly answers questions.
Arithmetics, GSM8K	Make sure to state your final answer in curly brackets at the very end of your response, just like: `{final answer: 12.34}`. Let's think step by step.
Form.Log., MMLU-Pro, BBH-Date	Make sure to state your final answer choice in curly brackets at the very end of your response, just like: `{final answer: (A)}`. Let's think step by step.
BBH-Navigate	Make sure to state your final answer choice in curly brackets at the very end of your response, just like: `{final answer: Yes}`. Let's think step by step.

Table 7: Overview of evaluation benchmarks, including the number of samples and representative examples.

Dataset	#Samples	Question	Answer
SciQ	1000	Compounds that are capable of accepting electrons, such as o 2 or f2, are called what?	oxidants
GPQA	198	Which of the following physical theories never requires UV regularization?	Superstring Theory
Arithmetics	200	What is $23 + 47$?	70
GSM8K	200	A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?	3
Formal Logic	126	Select the best translation into predicate logic: Sheena is a punk rocker. *Choices:* ("Sx", "xS", "sP", "Ps")	"Ps"
MMLU-Pro	105	Determine the number of men needed to build a boat in 77 days if it takes 36 men 132 days to build one. *Choices:* ("84 men", "36 men", "99 men", "132 men", "45 men", "70 men", "62 men", "50 men", "77 men", "120 men")	"62 men"
BBH-Date	250	Today is Christmas Eve of 1937. What is the date tomorrow in MM/DD/YYYY? *Options:* (A) 12/11/1937 (B) 12/25/1937 (C) 01/04/1938 (D) 12/04/1937 (E) 12/25/2006 (F) 07/25/1937	(B)
BBH-Navigate	250	If you follow these instructions, do you return to the starting point? Always face forward. Take 1 step backward. Take 9 steps left. Take 2 steps backward. Take 6 steps forward. Take 4 steps forward. Take 4 steps backward. Take 3 steps right. *Options:* - Yes - No	No

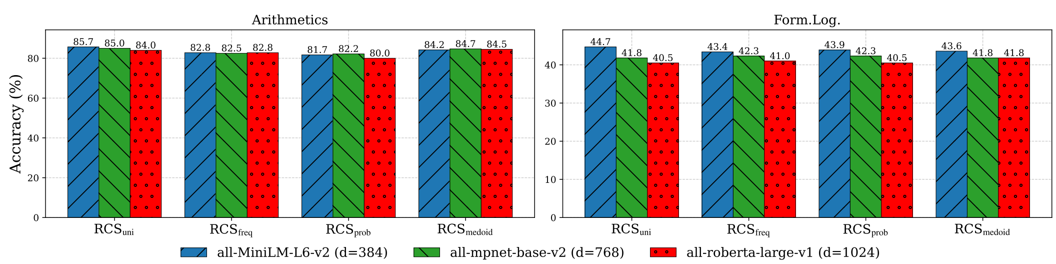

Figure 6: Effect of the sentence embedding model on Arithmetics and Form.Log. using Qwen2.5-3B.

A.7 Extended Results: Number Of Samples

Tables 9, 10, and 11 report performance over different number of sampling responses.

Table 8: Best-of-N selection accuracy across settings on Qwen2.5-7B and Gemma2-9B. **BBF** indicates whether the method is **black-box friendly**. For non-RCS methods, **bold** denotes statistically significant improvement over the best RCS variant ($p < 0.05$, paired t-test). For RCS methods (shaded), **bold** highlights the best-performing RCS variant per setting.

Model	Method	BBF	SciQ	GPQA	Arithmetics	GSM8K	Form.Log.	Average
Qwen2.5-7B	Greedy	✓	64.1	20.5	97.0	88.3	43.6	63.1
	Oracle	✓	82.0±1.1	49.9±1.6	100.0±0.0	94.6±0.8	67.7±2.5	78.8
	NLL	✗	66.4±0.1	18.3±1.3	47.3±5.7	32.7±4.3	25.7±0.5	38.1
	ANLL	✗	66.2±0.4	18.1±0.9	47.0±5.6	32.7±4.4	25.9±0.5	38.0
	SC	✓	**70.3±0.9**	22.1±4.2	74.3±2.0	52.3±2.0	**48.4±0.0**	53.5
	CE	✗	**70.4±0.4**	21.6±3.8	**75.5±4.0**	51.9±3.1	**48.7±1.2**	53.6
	ModeX	✓	69.0±0.6	21.6±1.6	54.3±5.1	37.6±3.3	35.2±2.6	43.5
	SCW	✓	70.2±0.8	26.3±3.2	74.8±4.7	56.3±2.3	48.1±1.2	55.1
	$\text{RCS}_{\text{medoid}}$	✓	70.1±0.6	25.0±1.3	77.5±3.6	57.5±2.6	**49.2±2.9**	55.9
	RCS_{uni}	✓	70.2±0.2	**27.1±1.7**	77.5±2.3	61.6±2.8	**49.2±3.2**	57.1
	RCS_{freq}	✓	**70.3±0.5**	25.9±1.8	76.7±1.4	56.0±2.1	47.9±2.4	55.4
	$\text{RCS}_{\text{cosine}}$	✓	70.1±0.5	26.9±1.8	76.8±3.4	61.3±3.1	**49.2±3.2**	56.9
	RCS_{prob}	✗	70.0±0.4	**27.1±2.4**	**77.7±1.9**	**62.1±3.1**	**49.2±3.2**	**57.2**
Gemma2-9B	Greedy	✓	67.6	22.6	96.0	69.0	52.4	62.0
	Oracle	✓	84.6±0.6	48.5±4.2	98.2±0.6	95.3±0.3	77.8±5.7	80.9
	NLL	✗	72.9±1.1	**25.6±1.1**	94.8±1.3	67.9±1.1	48.1±2.8	61.9
	ANLL	✗	66.3±0.9	21.2±2.3	96.0±1.0	**73.1±0.9**	50.0±0.8	61.3
	SC	✓	**73.9±0.2**	19.7±0.9	**97.3±0.3**	**72.9±3.5**	**51.6±0.8**	63.1
	CE	✗	**73.6±0.6**	18.6±0.5	**97.5±0.0**	**73.1±3.7**	**52.1±1.7**	63.0
	ModeX	✓	71.7±0.2	19.5±2.2	92.7±2.8	62.6±1.5	**49.2±4.2**	59.1
	SCW	✓	74.0±0.6	25.4±0.5	97.2±0.6	71.6±4.4	51.8±1.7	64.0
	$\text{RCS}_{\text{medoid}}$	✓	**74.2±0.4**	24.4±1.4	97.2±0.6	72.9±3.5	52.6±0.9	64.3
	RCS_{uni}	✓	74.0±0.4	25.9±0.5	97.2±0.6	73.8±2.8	**52.9±2.0**	64.8
	RCS_{freq}	✓	73.9±0.5	25.7±0.6	**97.3±0.3**	72.6±3.7	51.8±1.2	64.3
	$\text{RCS}_{\text{cosine}}$	✓	74.1±0.4	25.7±0.3	97.2±0.6	71.1±4.5	51.8±1.2	64.0
	RCS_{prob}	✗	**74.2±0.5**	**26.3±0.6**	**97.3±0.3**	**75.1±3.5**	52.4±0.8	**65.1**

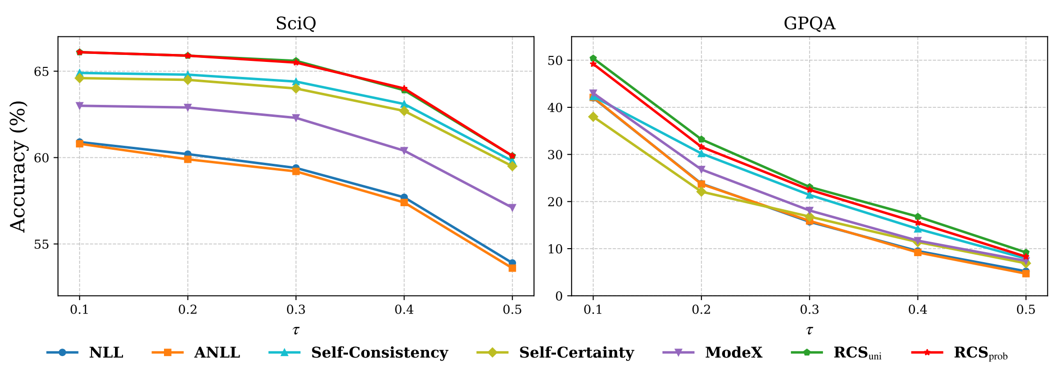

Figure 7: Performance on SciQ and GPQA when varying correctness threshold (τ) using Llama3.2-3B.

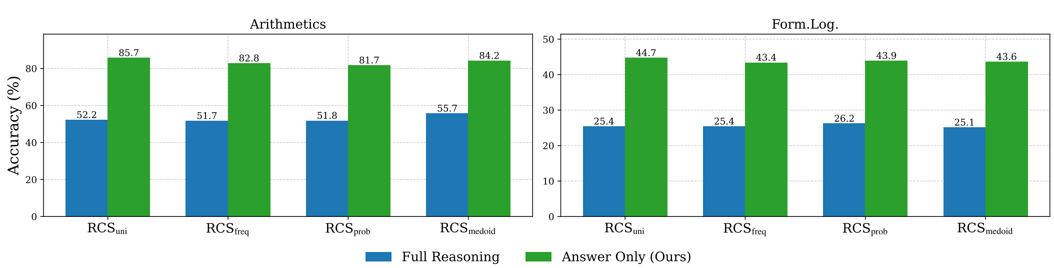

Figure 8: Effect of reasoning paths on Arithmetics and Form.Log. using Qwen2.5-3B.

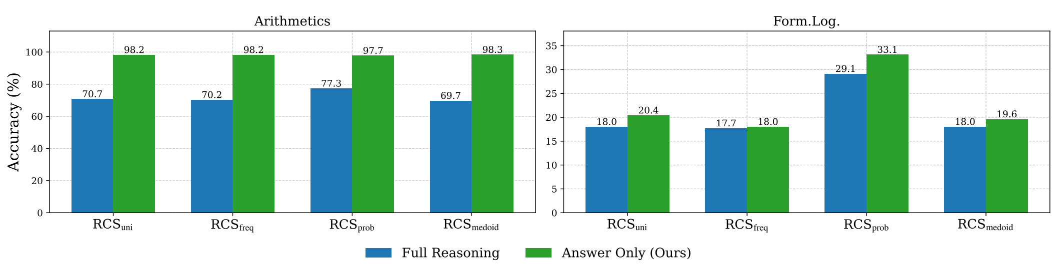

Figure 9: Effect of reasoning paths on Arithmetics and Form.Log. using Llama3.2-3B.

Table 9: Best-of-N selection accuracy across settings (N=5).

Model	Method	SciQ	GPQA	Arithmetics	GSM8K	Form.Log.
Qwen2.5-3B	NLL	61.7±1.2	17.6±3.1	49.2±12.4	45.2±4.7	28.0±3.2
	ANLL	61.6±1.1	17.1±4.1	48.7±11.4	46.0±4.8	27.5±3.3
	SC	63.8±0.5	18.1±2.7	78.7±7.2	65.0±0.5	39.4±1.8
	CE	63.6±0.3	15.7±1.7	78.5±7.4	66.8±5.0	42.4±0.2
	ModeX	61.2±1.2	18.0±2.4	46.0±3.0	32.0±0.5	24.9±0.9
	SCW	64.0±0.2	24.0±2.6	72.2±2.5	55.8±2.2	36.8±1.2
	$\text{RCS}_{\text{medoid}}$	64.0±0.4	24.7±2.2	77.7±2.8	66.7±2.9	40.7±0.5
	RCS_{uni}	63.8±0.1	24.2±2.9	77.0±2.8	66.5±4.0	40.7±0.5
	RCS_{freq}	64.0±0.2	24.0±2.6	77.5±3.1	66.3±3.8	41.0±0.5
	$\text{RCS}_{\text{cosine}}$	63.9±0.2	24.9±2.4	72.8±2.5	55.5±2.4	36.5±1.4
	RCS_{prob}	64.0±0.5	21.6±2.0	75.7±5.3	63.1±4.4	41.3±0.0
Qwen2.5-7B	NLL	67.6±1.5	17.1±1.4	63.3±10.9	51.4±3.4	36.0±2.3
	ANLL	67.5±1.1	16.9±1.2	63.5±11.0	51.4±3.4	35.4±2.0
	SC	69.2±0.5	22.3±3.2	83.3±13.7	72.4±3.1	47.4±4.4
	CE	69.4±0.1	19.9±1.1	84.0±16.0	74.3±2.6	47.8±3.6
	ModeX	69.0±0.7	22.1±3.0	54.2±3.2	40.6±2.7	34.1±6.3
	SCW	68.9±0.4	26.9±0.9	74.8±4.6	58.2±4.4	45.0±4.0
	$\text{RCS}_{\text{medoid}}$	69.0±0.4	27.3±0.8	84.3±12.0	74.3±1.8	47.6±3.6
	RCS_{uni}	68.9±0.4	27.6±1.3	84.2±11.8	75.0±1.3	47.6±3.6
	RCS_{freq}	69.0±0.4	26.9±0.9	84.3±12.0	75.1±1.0	47.6±3.6
	$\text{RCS}_{\text{cosine}}$	69.0±0.4	27.5±1.0	74.8±6.0	60.1±2.4	45.2±4.4
	RCS_{prob}	69.3±0.4	25.2±1.5	83.0±11.8	70.9±2.0	47.9±3.3
Llama3.2-3B	NLL	54.0±1.3	19.2±0.9	82.7±4.2	73.4±3.8	31.0±2.9
	ANLL	53.7±0.6	18.5±2.0	86.7±3.3	76.1±1.8	31.5±2.0
	SC	58.6±0.8	16.1±2.4	93.3±2.1	79.2±3.6	31.5±1.7
	CE	57.6±1.4	19.2±3.2	93.0±1.8	80.3±4.1	33.7±2.4
	ModeX	53.4±1.1	12.8±3.2	68.8±1.0	61.3±2.6	19.1±3.5
	SCW	59.4±1.2	22.3±2.6	90.5±2.1	74.3±0.3	22.0±5.4
	$\text{RCS}_{\text{medoid}}$	59.4±1.2	22.5±2.9	94.0±0.5	80.2±2.0	30.9±4.8
	RCS_{uni}	59.3±1.5	22.3±2.6	93.3±0.6	80.4±2.1	30.9±4.8
	RCS_{freq}	59.4±1.2	22.3±2.6	93.8±0.8	80.4±2.1	30.9±4.8
	$\text{RCS}_{\text{cosine}}$	59.3±1.4	22.5±2.9	92.8±1.1	76.3±1.6	20.9±4.4
	RCS_{prob}	58.9±1.4	21.2±3.6	93.5±1.5	80.0±1.9	32.8±2.4
Llama3.1-8B	NLL	61.2±0.6	21.1±1.1	73.3±2.5	72.8±3.3	38.9±1.4
	ANLL	60.8±0.5	20.7±1.4	73.2±3.3	73.3±3.1	38.4±1.8
	SC	66.0±1.8	19.2±1.9	82.2±7.2	82.6±3.8	43.1±3.9
	CE	65.6±0.4	19.7±3.2	82.1±8.4	82.4±3.1	42.4±3.0
	ModeX	62.1±0.9	20.5±2.1	50.2±0.8	53.5±3.8	25.1±0.5
	SCW	67.1±0.7	26.8±1.3	79.0±0.5	69.5±1.3	34.7±0.9
	$\text{RCS}_{\text{medoid}}$	67.0±0.8	27.3±1.3	86.0±4.0	83.2±5.2	41.8±3.2
	RCS_{uni}	67.0±0.6	26.9±2.1	85.5±3.0	83.2±5.6	42.3±3.6
	RCS_{freq}	67.1±0.7	26.8±1.3	85.7±3.6	83.2±5.6	42.3±3.6
	$\text{RCS}_{\text{cosine}}$	66.9±0.6	27.1±2.1	81.3±0.8	70.9±2.4	36.8±2.6
	RCS_{prob}	66.7±0.8	26.4±2.7	77.3±3.6	76.0±3.1	39.2±4.1
Gemma2-9B	NLL	69.2±0.4	27.1±1.1	94.5±1.0	87.3±0.9	54.5±4.4
	ANLL	66.2±0.9	21.1±1.1	95.8±1.0	87.8±1.3	54.8±2.1
	SC	72.8±0.6	17.3±2.3	97.0±0.9	88.8±1.8	55.6±5.0
	CE	73.2±0.4	17.8±3.5	97.0±0.9	89.8±2.1	55.5±5.2
	ModeX	71.5±0.6	20.7±0.5	92.5±1.0	64.0±2.3	52.1±6.5
	SCW	73.9±0.6	24.9±2.3	96.7±0.8	72.1±0.9	54.0±2.9
	$\text{RCS}_{\text{medoid}}$	73.7±0.7	24.2±2.0	96.8±0.6	89.0±2.1	55.3±4.8
	RCS_{uni}	73.8±0.4	25.9±2.1	96.8±0.6	88.7±1.6	55.8±4.8
	RCS_{freq}	74.0±0.6	24.9±2.3	96.8±0.6	88.7±1.6	55.6±4.4
	$\text{RCS}_{\text{cosine}}$	73.8±0.5	25.7±2.2	96.8±0.6	72.8±1.1	54.0±4.2
	RCS_{prob}	73.2±0.3	24.2±1.3	96.8±0.6	88.8±1.5	55.3±3.6

Table 10: Best-of-N selection accuracy across settings (N=20).

Model	**Method**	**SciQ**	**GPQA**	**Arithmetics**	**GSM8K**	**Form.Log.**
Qwen2.5-3B	NLL	59.0±0.4	16.4±2.3	48.2±11.9	40.6±0.5	26.5±2.6
	ANLL	58.4±0.4	16.2±2.6	47.8±12.5	40.8±1.1	26.5±2.6
	SC	64.9±0.6	20.0±0.3	95.3±5.1	88.3±0.9	48.1±3.6
	CE	64.2±0.9	18.3±1.3	95.3±5.9	88.8±0.9	47.1±2.4
	ModeX	62.2±0.8	21.9±1.7	51.7±3.4	37.9±2.1	23.3±1.8
	SCW	64.9±0.6	24.4±2.7	92.5±0.7	67.0±1.3	42.1±2.9
	$\text{RCS}_{\text{medoid}}$	65.0±0.7	26.1±2.7	96.5±3.5	88.5±0.6	47.4±4.1
	RCS_{uni}	65.5±1.0	24.9±2.7	96.7±2.4	87.1±1.6	48.1±4.4
	RCS_{freq}	64.7±0.5	23.8±2.9	96.3±3.3	88.0±0.8	47.6±5.0
	$\text{RCS}_{\text{cosine}}$	65.3±0.9	25.2±2.9	94.2±1.1	70.4±1.8	42.3±3.9
	RCS_{prob}	65.3±0.7	24.9±2.4	95.2±3.3	86.6±1.1	46.6±4.1
Qwen2.5-7B	NLL	66.5±0.8	18.1±1.9	63.3±11.2	55.8±2.0	30.7±4.8
	ANLL	66.5±1.0	18.1±1.8	63.5±12.2	55.7±1.8	31.2±4.4
	SC	70.0±0.2	26.3±0.6	93.3±9.8	93.1±1.3	51.8±2.3
	CE	69.9±0.2	25.2±0.3	93.7±9.7	92.6±0.8	52.9±2.6
	ModeX	68.5±1.2	24.0±2.6	59.5±0.0	37.9±1.5	31.2±3.6
	SCW	70.0±0.3	27.6±2.0	80.8±0.4	67.0±1.3	48.7±1.8
	$\text{RCS}_{\text{medoid}}$	69.7±0.3	28.0±1.8	95.0±6.9	92.2±0.8	52.6±1.7
	RCS_{uni}	69.9±0.2	28.7±2.2	95.8±5.9	91.7±1.1	53.4±2.4
	RCS_{freq}	70.1±0.3	28.0±1.6	94.5±7.8	92.7±1.1	52.1±1.8
	$\text{RCS}_{\text{cosine}}$	69.8±0.2	28.3±1.8	86.0±2.8	70.4±2.4	48.7±1.7
	RCS_{prob}	70.0±0.3	27.5±1.6	95.3±6.8	91.7±0.3	53.4±3.2
Llama3.2-3B	NLL	53.0±0.5	22.1±2.9	90.2±0.8	79.9±2.1	39.7±0.0
	ANLL	52.8±0.9	19.3±3.5	91.7±0.3	82.7±0.9	39.9±5.3
	SC	59.8±0.3	15.7±0.8	98.2±0.3	89.3±1.3	43.7±2.9
	CE	60.0±0.6	17.3±1.5	98.2±0.3	89.7±1.3	45.0±3.0
	ModeX	55.0±0.7	16.9±3.2	73.0±0.5	61.8±5.7	20.1±2.0
	SCW	60.3±0.4	25.0±2.6	97.5±nan	84.1±1.3	24.6±2.4
	$\text{RCS}_{\text{medoid}}$	60.7±0.2	26.1±3.4	98.0±0.0	89.5±1.3	43.9±3.2
	RCS_{uni}	60.7±0.7	26.8±2.9	98.5±0.5	89.5±1.6	44.2±2.4
	RCS_{freq}	60.4±0.3	25.7±2.9	98.0±0.0	89.3±1.3	43.4±2.5
	$\text{RCS}_{\text{cosine}}$	60.8±0.6	26.1±4.0	98.5±nan	84.3±1.5	22.8±1.7
	RCS_{prob}	60.7±0.5	26.4±3.7	98.0±0.0	89.5±1.1	42.1±2.4
Llama3.1-8B	NLL	61.0±1.5	20.7±1.5	81.2±4.3	83.8±1.8	48.9±1.8
	ANLL	61.1±1.9	21.8±0.0	79.8±3.9	82.2±3.1	44.4±2.1
	SC	68.1±0.5	18.9±0.4	95.3±6.0	91.9±1.3	55.3±2.0
	CE	68.2±0.6	20.7±0.0	94.8±7.2	91.7±1.5	54.0±2.1
	ModeX	62.8±0.7	22.3±0.5	54.8±0.6	49.9±3.6	24.3±3.3
	SCW	68.3±0.1	32.4±0.4	92.3±0.8	84.3±1.3	39.7±3.5
	$\text{RCS}_{\text{medoid}}$	67.9±0.1	35.8±0.7	97.8±2.1	92.2±1.8	55.0±2.4
	RCS_{uni}	68.3±0.5	36.0±1.1	98.8±1.2	91.4±1.8	54.0±2.4
	RCS_{freq}	68.3±0.5	33.7±0.7	96.0±5.2	92.2±1.8	55.0±2.4
	$\text{RCS}_{\text{cosine}}$	68.3±0.3	36.0±2.6	95.3±1.4	83.6±1.1	39.4±3.7
	RCS_{prob}	68.1±0.3	36.3±0.7	97.2±1.2	88.3±1.8	52.9±3.0
Gemma2-9B	NLL	74.1±0.3	27.1±1.1	96.2±0.3	89.3±0.9	52.1±0.9
	ANLL	64.8±0.5	23.0±1.1	96.3±0.3	89.8±0.9	54.5±1.7
	SC	74.1±0.8	22.1±2.0	97.2±0.3	91.7±0.3	54.0±0.8
	CE	74.3±0.4	21.8±0.5	97.2±0.3	91.7±0.8	54.0±0.8
	ModeX	73.0±0.4	22.1±1.6	92.3±2.0	64.8±1.1	51.9±5.4
	SCW	74.1±0.5	28.8±2.0	97.0±0.5	75.8±0.3	51.3±0.5
	$\text{RCS}_{\text{medoid}}$	74.2±0.4	28.7±1.7	97.2±0.3	91.7±0.3	54.2±0.5
	RCS_{uni}	74.7±0.4	29.9±1.3	97.2±0.3	91.7±0.3	54.2±0.5
	RCS_{freq}	74.2±0.6	29.4±2.1	97.2±0.3	91.9±0.0	54.2±0.5
	$\text{RCS}_{\text{cosine}}$	74.6±0.4	29.0±0.5	97.2±0.3	76.3±0.8	52.1±0.9
	RCS_{prob}	74.6±0.6	29.5±0.9	97.2±0.3	91.5±0.3	54.2±0.9

Table 11: Best-of-N selection accuracy across settings (N=40).

Model	Method	SciQ	GPQA	Arithmetics	GSM8K	Form.Log.
Qwen2.5-3B	NLL	56.3±0.8	16.8±1.8	45.3±13.0	42.8±1.5	25.9±3.8
	ANLL	56.1±0.9	16.8±2.0	44.8±13.3	43.0±2.8	25.1±3.8
	SC	65.2±0.4	21.8±1.4	97.3±2.9	90.2±1.1	49.5±1.2
	CE	65.2±0.3	20.4±2.7	97.5±2.6	90.4±0.5	48.9±3.2
	ModeX	63.0±1.0	20.0±0.5	49.8±3.2	35.7±3.1	25.7±6.7
	SCW	65.3±0.3	25.6±2.3	89.0±2.1	71.7±1.2	44.2±1.7
	$\text{RCS}_{\text{medoid}}$	65.5±0.5	25.0±1.3	98.2±1.5	90.0±0.8	47.9±1.2
	RCS_{uni}	65.7±0.7	25.0±1.3	98.5±1.3	89.7±1.3	48.4±0.8
	RCS_{freq}	65.3±0.5	23.8±2.4	97.7±2.4	90.0±0.8	47.6±2.1
	$\text{RCS}_{\text{cosine}}$	65.7±0.7	25.7±0.6	97.2±1.1	76.1±2.2	44.7±1.2
	RCS_{prob}	65.9±0.7	24.4±1.0	98.2±1.6	89.3±1.3	47.4±0.5
Qwen2.5-7B	NLL	66.0±0.4	18.5±1.2	63.0±10.1	57.5±3.1	30.7±4.4
	ANLL	65.9±0.6	18.5±0.8	62.8±9.3	57.5±3.4	30.4±3.2
	SC	70.5±0.3	26.1±1.6	96.2±5.8	94.4±0.5	54.0±0.8
	CE	70.4±0.6	26.9±3.1	96.3±5.5	94.4±0.0	53.7±1.2
	ModeX	69.6±0.3	25.0±1.7	53.2±1.8	38.6±1.8	31.2±2.6
	SCW	70.4±0.1	28.2±0.6	86.0±0.0	74.6±2.8	49.5±2.4
	$\text{RCS}_{\text{medoid}}$	70.0±0.1	28.0±0.9	97.2±4.0	94.8±0.3	53.2±2.1
	RCS_{uni}	70.3±0.4	29.0±0.9	98.2±2.3	94.6±0.3	52.6±3.7
	RCS_{freq}	70.5±0.2	27.5±0.5	96.5±5.2	94.6±0.3	52.9±3.2
	$\text{RCS}_{\text{cosine}}$	70.2±0.4	28.3±0.3	94.0±0.0	75.6±1.8	50.3±1.7
	RCS_{prob}	70.4±0.4	27.3±0.8	97.5±3.5	94.6±0.3	52.4±3.5
Llama3.2-3B	NLL	53.4±0.3	21.6±1.7	93.5±2.6	82.7±1.0	36.2±4.0
	ANLL	52.8±0.6	18.7±2.3	93.3±2.5	83.2±1.8	41.8±5.7
	SC	60.4±0.7	17.6±0.9	98.7±0.6	91.2±0.8	44.2±0.5
	CE	60.2±0.4	21.2±3.2	98.8±0.3	91.2±0.6	43.1±0.9
	ModeX	55.8±0.9	19.0±2.9	72.5±2.6	57.0±4.1	22.0±3.2
	SCW	60.6±0.9	24.7±2.4	98.8±0.4	86.6±1.1	22.2±4.2
	$\text{RCS}_{\text{medoid}}$	60.9±1.0	29.4±1.6	98.7±0.6	91.2±0.6	45.2±0.8
	RCS_{uni}	60.9±0.8	29.4±2.0	98.5±0.9	90.4±0.5	46.3±1.7
	RCS_{freq}	61.0±1.0	26.9±1.9	98.7±0.6	90.9±0.5	44.2±1.7
	$\text{RCS}_{\text{cosine}}$	61.1±1.1	29.9±2.0	98.8±1.1	86.3±0.5	23.0±2.9
	RCS_{prob}	61.0±0.6	29.4±2.1	98.3±0.3	90.7±0.3	44.7±2.8
Llama3.1-8B	NLL	61.8±0.9	18.9±1.1	83.7±1.2	83.9±2.5	46.8±2.1
	ANLL	61.3±0.8	20.5±1.8	84.0±1.7	81.7±2.0	43.4±4.0
	SC	69.3±0.3	19.7±0.7	96.3±4.2	93.2±0.8	51.9±2.0
	CE	68.5±0.4	18.9±0.4	96.5±4.3	93.2±0.8	51.1±1.7
	ModeX	64.8±0.3	19.3±2.7	53.5±0.9	51.4±1.6	23.0±2.1
	SCW	69.2±0.4	25.6±1.1	98.8±0.8	85.8±0.5	37.6±1.2
	$\text{RCS}_{\text{medoid}}$	69.1±0.3	31.6±0.7	98.5±0.5	92.4±0.5	51.1±1.2
	RCS_{uni}	69.6±0.2	31.4±0.4	98.7±1.0	92.2±0.3	50.5±1.2
	RCS_{freq}	69.2±0.8	28.2±1.8	96.7±3.6	93.1±0.6	51.1±1.2
	$\text{RCS}_{\text{cosine}}$	69.5±0.2	31.9±1.1	99.2±0.6	84.9±0.8	38.1±1.4
	RCS_{prob}	69.6±0.4	29.5±1.5	98.5±0.9	91.9±1.3	51.6±2.1
Gemma2-9B	NLL	74.2±0.1	26.8±2.1	96.0±0.5	89.7±0.8	51.3±3.9
	ANLL	62.2±1.5	20.7±1.6	96.2±0.3	89.5±1.8	53.7±2.0
	SC	74.2±0.4	22.5±0.8	97.5±0.0	91.7±0.8	55.6±0.8
	CE	74.2±0.3	19.3±0.6	97.5±0.0	91.5±0.3	55.3±1.2
	ModeX	71.9±0.4	21.4±3.2	93.5±0.5	66.3±2.9	54.0±2.4
	SCW	74.2±0.5	26.9±0.9	97.5±0.0	77.2±0.9	52.4±0.8
	$\text{RCS}_{\text{medoid}}$	74.4±0.4	29.4±1.2	97.5±0.0	91.7±0.8	55.0±0.9
	RCS_{uni}	74.4±0.8	29.7±1.6	97.5±0.0	91.5±0.6	54.8±0.8
	RCS_{freq}	74.2±0.4	28.0±1.0	97.5±0.0	91.7±0.8	55.3±1.2
	$\text{RCS}_{\text{cosine}}$	74.3±0.6	29.9±0.8	97.5±0.0	76.7±0.9	52.6±0.5
	RCS_{prob}	74.4±0.5	29.7±0.8	97.3±0.3	91.2±0.8	55.3±0.5

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